

User Guide - SonarBun Live Translator

Live translation for browser audio, desktop audio, microphones, and virtual audio devices.

1. What it does

SonarBun Live Translator listens to an audio source, transcribes speech with Whisper, and, when configured, translates the transcript through online API services or through a local offline model. It is designed for livestreams, VODs, YouTube videos, Twitch streams, meetings, and foreign-language content.

The app displays the source and translated text in the main window and updates live.txt with the latest translation, which can be used by OBS overlays or other tools.

2. Starting the app

- 1 Open SonarBunLiveTranslator.exe.
- 2 Choose the audio source: **Output / browser audio** or **Input / microphone**.
- 3 Select the correct audio device.
- 4 Select source and target languages.
- 5 In **Settings**, choose **Online** to use an online API service or **Offline** to use a local model.
- 6 In Online mode, configure provider and API key. In Offline mode, download the local model with the dedicated button.
- 7 Press **Start**.

3. Live tab

Field	Description
Source	Chooses whether to listen to system/browser output or to a physical/virtual input.
Device	The audio device to listen to. Output shows loopback devices; Input shows microphones and virtual inputs.
Show all input APIs	Shows every audio backend. By default, the app keeps only MME and Windows WASAPI because they are more reliable for this pipeline.
From	The spoken language. Choosing a specific language, such as Japanese or English, reduces recognition errors.
To	The language to translate into.
Start / Stop	Starts or stops listening.

4. Input or Output?

Output / browser audio

Use this mode for audio coming from a browser, YouTube, Twitch, a video player, or desktop audio. The app captures audio through WASAPI loopback using PyAudioWPatch.

Input / microphone

Use this mode for microphones, audio interfaces, or other virtual inputs. The normal list shows MME and Windows WASAPI. DirectSound is hidden by default because it is often listed by Windows but may be less reliable with this app.

5. Settings

Setting	Purpose
UI language	Changes the app interface language: English or Japanese. It does not change translation languages.
Translation mode	Online uses an online API service. Offline uses a local model stored in the app folder.
Translation provider	Chooses the LLM provider used for online translation.
Translation model	The model used to translate the transcript.
API key	Your personal provider key. It is encrypted with Windows DPAPI and is not stored as plain text.
Offline model	The local model used when Translation mode is Offline. Current option: NLLB-200 distilled 600M CT2 INT8.
Model status	Shows whether the offline model is installed. If it is missing, the download button is enabled.
Download model	Downloads the selected offline model. Internet is required only for the first download.
Offline device	cpu or cuda for offline translation.
Offline compute	Compute type for the offline model. Recommended default: int8.
Glossary profile	Optional glossary profile. None disables it. It helps manage recurring terms, proper names, slang, item names, and context-specific words that should not be translated too literally.
Whisper model	The faster-whisper transcription model. small is fast; medium is more accurate but slower.
Whisper device	cpu or cuda. If you have a compatible NVIDIA GPU, cuda is recommended.
Compute type	int8 for CPU; float16 for CUDA; float32 only when needed.
Beam size	Quality vs speed. For live use, set it to 1.
VAD mode	Voice detection aggressiveness. 0 is permissive, 3 is strict. Recommended default: 2.
Max segment seconds	Maximum audio chunk length before transcription. Lower values reduce latency but may split sentences.
Silence tail ms	How much silence is needed to close a sentence. 250-350 ms is a good starting point.
Live output file	The file updated with the latest translation. Default: live.txt.

6. API providers

The providers in this section are the currently supported options or practical examples. More online API services may be added in the future if needed.

OpenAI

OpenAI is recommended when translation quality matters most, especially for Japanese, context, slang, tone, and ambiguous sentences. It requires a personal API key and usage is billed according to the account.

Groq

Groq is useful for fast, low-cost translation and testing. Use the official console: <https://console.groq.com/home>.

Groq provides free/rate-limited access for getting started. It is free because Groq offers a developer trial/free tier with usage limits and rate limits. If you exceed those limits, you must wait for reset or upgrade to a paid/developer plan. Limits can change, so always check the official console and pricing page.

To get a Groq key:

- 1 Go to <https://console.groq.com/home>.
- 2 Sign in or create an account.
- 3 Open API Keys / Keys.
- 4 Create a new API key.
- 5 Copy it into SonarBun after selecting provider **Groq**.
- 6 Press **Save Settings**, then **Test**.

Never share your API key. If it is exposed, revoke it in the provider console and create a new one.

7. Offline mode

Offline mode translates without API services and without an internet connection, except for downloading the local model the first time. Audio is transcribed locally with faster-whisper, then translated with a local NLLB model in CTranslate2 format.

The first use requires downloading the selected model with **Download model**. The current model is about 630 MB. After the download, normal offline translation does not send sentences to external services.

Offline mode is more private, but it can be less natural and less context-aware than online API services. On CPU it can also be slower; with an NVIDIA GPU and cuda, performance can improve.

8. Glossary and cache

Glossary profile selects an optional term profile. It is meant to protect item names, proper names, slang, and game-specific terms that generic models may translate too literally.

In **Online** mode, relevant glossary entries are sent to online API services as supporting translation instructions. In **Offline** mode, NLLB does not accept contextual instructions, so the glossary is used as a local pre/post-translation correction layer.

The file is glossaries/glossary.json. Users can maintain it by following the same internal schema. Restart the app after editing it.

SonarBun also uses a local cache in translation_cache.json. If the same sentence is translated again with the same engine, languages, and glossary, the saved translation is reused. This reduces latency, API calls, and variation between identical translations.

9. Generated files

File/folder	Description
live.txt	Latest translation. Rewritten whenever a useful phrase is processed.
history.txt	Session history with source text and translation.
sonarbun_config.json	Non-sensitive settings: languages, device, model, provider, audio parameters.
models/	Cache for downloaded faster-whisper models.
models/local_translation/	Cache for offline translation models downloaded with Download model.
glossaries/glossary.json	Editable glossary profiles.

File/folder	Description
translation_cache.json	Local cache of previously calculated translations.
AppData/Local/SonarBunLiveTranslator/*.dpapi	API keys encrypted with Windows DPAPI.

10. Setup CUDA on Windows

CUDA is only needed if you want to use an NVIDIA GPU for Whisper or offline translation. If you are not sure, use `cpu + int8`: it is slower, but it does not require extra NVIDIA setup.

To use `cuda`, the PC needs a compatible NVIDIA GPU, updated NVIDIA drivers, and a working CUDA/cuDNN environment. The driver alone may not be enough, because some components look for DLLs such as `cuBLAS`, `CUDA runtime`, and `cuDNN`.

Recommended installation

- 1 Check that the PC has a compatible NVIDIA GPU.
- 2 Install or update the official NVIDIA driver for that GPU.
- 3 Restart Windows.
- 4 Open Command Prompt and run `nvidia-smi`. If it shows the GPU, the driver is active.
- 5 If SonarBun fails when using `cuda`, install CUDA Toolkit for Windows from NVIDIA's official page.
- 6 If the error mentions `cuDNN` or missing DLLs, install `cuDNN` following NVIDIA's official guide.
- 7 Restart Windows and try SonarBun again with `cuda + float16`.

Note: the CUDA version shown by `nvidia-smi` is the maximum CUDA version supported by the driver, not necessarily the installed Toolkit version.

CUDA troubleshooting table

Situation	Likely cause	Fix
No NVIDIA GPU	Intel/AMD GPU or integrated graphics only	Use <code>cpu</code> and <code>int8</code> .
<code>nvidia-smi</code> is not found	NVIDIA driver missing or <code>PATH</code> not refreshed	Install/update the NVIDIA driver and restart.
<code>nvidia-smi</code> works but SonarBun fails on CUDA	CUDA/cuBLAS/cuDNN runtime required by <code>CTranslate2</code> is missing	Install CUDA Toolkit and, if needed, <code>cuDNN</code> .
Error mentions <code>cuda</code> or <code>cuBLAS</code> DLL missing	NVIDIA DLLs are not found	Install compatible <code>cuDNN/CUDA</code> or make sure the bin folders are in <code>PATH</code> .
CUDA starts but is unstable	Old driver, low VRAM, or model too heavy	Update the driver, use a smaller Whisper model, or go back to <code>CPU/int8</code> .
Non-technical user	CUDA setup is too complex	Keep the default <code>cpu + int8</code> .

11. Recommended settings

Fast

Whisper small, `cuda`, `float16`, beam 1, max segment 2, silence tail 250-300, fast online provider.

Balanced

Whisper medium, `cuda`, `float16`, beam 1, max segment 2-4, online provider depending on preference.

CPU

Whisper small or base, `cpu`, `int8`. Latency will be higher.

Offline

Translation mode: Offline, model NLLB-200 distilled 600M CT2 INT8, Offline compute: int8. Use cuda if available, otherwise cpu.

12. Troubleshooting

Problem	Try this
No audio	Change device, check Output/Input mode, try WASAPI or MME, verify Windows volume.
Missing translations	Check API key, provider, internet connection, and history.txt.
High latency	Use a smaller Whisper model, CUDA, beam 1, shorter segments, or a faster provider.
API test fails	Check that the key is valid and the provider has available quota.
Offline model download fails	Check the connection and press Download model again. The download can reuse partially downloaded files.
Offline mode does not translate	Check that Model status says Installed and that the selected language pair is supported.
Glossary is not applied	Check that the profile is selected in Settings and restart the app after editing glossary.json.
Old or unexpected translation	The cache may reuse a saved translation. Delete translation_cache.json if you want a clean run after many glossary edits.
Bad translation	Set the source language manually instead of using Auto detect.

13. Useful links

- 1 Groq Console: <https://console.groq.com/home>
- 1 Groq Quickstart: <https://console.groq.com/docs/quickstart>
- 1 Groq Pricing: <https://groq.com/pricing>
- 1 NVIDIA CUDA Installation Guide for Windows: <https://docs.nvidia.com/cuda/cuda-installation-guide-microsoft-windows/>
- 1 NVIDIA cuDNN Installation Guide: <https://docs.nvidia.com/deeplearning/cudnn/installation/latest/>
- 1 CTranslate2 GPU documentation: <https://opennmt.net/CTranslate2/installation.html>

For any doubt, question, or problem to solve, contact me directly. X account: **@1pixeldot**.